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Factor Affecting Agroforestry System

Course - *Fundamental of Agroforestry*

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Introduction

Agroforestry System (AFS)

Agroforestry System is a sustainable land-use system that integrates trees, crops, and livestock on the same land to maximize productivity and environmental benefits. It combines elements of agriculture and forestry, promoting biodiversity, soil conservation, and efficient resource use.

Importance of Agroforestry System:

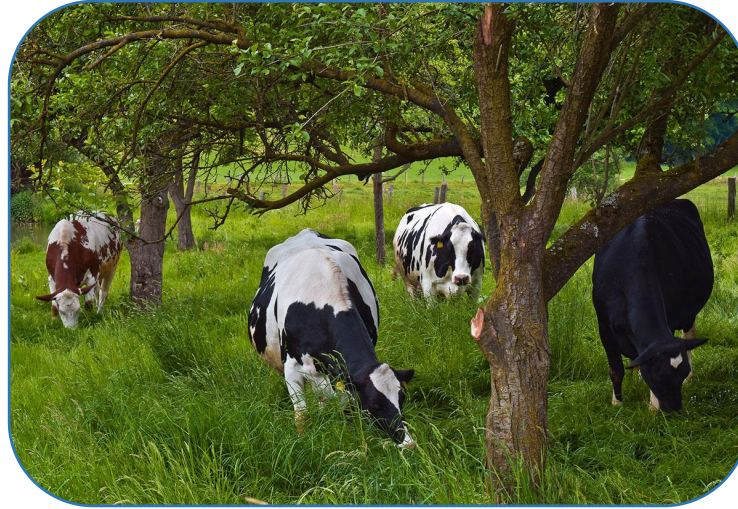
- Enhances soil fertility and prevents soil erosion.
- Improves water retention and reduces runoff.
- Provides food, fuelwood, fodder, and timber for farmers.
- Increases carbon sequestration and mitigates climate change.
- Supports livelihood diversification by generating multiple income sources.

Types of Agroforestry System



I) Agrisilvicultural Systems

These systems combine crops and trees, such as in alley cropping or home gardens, where trees and agricultural crops are cultivated together to enhance productivity and sustainability.



II) Silvopastoral Systems

This approach integrates trees with livestock grazing areas, providing shade and fodder for animals while improving land productivity. An example is the combination of trees and grazing in rangelands.



III) Agrosilvopastoral Systems

These systems incorporate trees, crops, and livestock on the same land, such as home gardens that include animals, promoting a holistic approach to land use.

Factors affecting AFS

1. Climatic Factors
2. BioPhysical Factors
3. Socio-economical Factor
4. Resource availability

Climatic Factors

➤ Rainfall:

- Determines water availability for crops and trees.
- Affects soil moisture and overall productivity.

➤ Solar Radiation:

- Influences photosynthesis and plant growth.
- Determines the selection of shade-tolerant or sun-loving species.

➤ Temperature:

- Affects plant metabolism, growth rates, and species adaptability.
- Extreme temperatures can cause heat stress or frost damage.

➤ Humidity:

- Impacts disease prevalence, especially fungal infections.
- Affects transpiration and water requirements of plants.

➤ Rainfall & Soil Erosion:

- Heavy rainfall increases **soil erosion and nutrient leaching**.
- Leads to sedimentation and surface runoff, affecting farm productivity.

Biophysical Factors

➤ Soil Characteristics:

- Soil fertility, texture, and organic matter influence tree and crop growth.
- Poor soils in tropical regions require **nutrient management strategies**.

➤ Slope and Terrain:

- Steep slopes increase **erosion risk** but can be managed with contour planting.
- Flat terrains are better for intensive agroforestry practices.

➤ Plant Communities:

- The presence of native vegetation affects biodiversity and ecological balance.
- Competition between trees and crops for water, sunlight, and nutrients.

➤ Slash-and-Burn Agriculture:

- Common in upland areas but reduces soil fertility and **increases erosion**.
- Sustainable alternatives like **agroforestry alley cropping** help maintain soil health.

Socio-economic Factors

➤ **Farmer's Economic Status:**

- Many upland farmers are resource-poor and depend on subsistence farming.
- Lack of capital limits investment in agroforestry practices.

➤ **Subsistence Farming & Food Security:**

- Agroforestry must balance food production with other economic benefits.
- Farmers require cash income for essential needs like healthcare and education.

➤ **Market Access & Infrastructure:**

- Poor market access reduces incentives for agroforestry adoption.
- Availability of roads, storage, and processing facilities affects profitability.

➤ **Government Policies & Support:**

- Land tenure security encourages long-term investment in agroforestry.
- Extension services and subsidies can improve adoption rates.

➤ **Community & Cultural Aspects:**

- Cultural beliefs and traditional farming methods can impact the willingness to adopt new agroforestry techniques

Resource Availability

➤ **Water Resources:**

- Availability depends on rainfall patterns and seasonal changes.
- Agroforestry can help conserve water by reducing runoff and improving infiltration.
- Climate-related events like monsoons and cyclones impact water availability.

➤ **Land and Forest Resources:**

- Deforestation and land degradation reduce soil fertility and biodiversity.
- Slash-and-burn agriculture contributes to erosion and nutrient loss.
- Sustainable forest management is crucial for long-term agroforestry success.

➤ **Human Resources:**

- The availability of labor influences the feasibility of managing agroforestry components.
- Farmers' knowledge and skills influence agroforestry adoption.
- Training and extension services improve understanding of best practices.
- Adapting farming systems to local conditions ensures sustainability.

➤ **Seedling Access**

- Reliable sources of quality seedlings are crucial for establishing agroforestry systems.

Conclusion

Agroforestry is a sustainable land-use system influenced by climate, biophysical, socioeconomic, and resource factors. Climate variables impact plant growth, while soil and landscape features determine productivity. Socioeconomic conditions affect adoption, and efficient resource management ensures long-term success. Integrated planning enhances productivity and biodiversity. Advancements in climate-resilient agroforestry, research, and policy support will strengthen its role in sustainable agriculture. Agroforestry offers environmental, economic, and social benefits, promoting food security, rural development, and ecological conservation.

THANK YOU

Do you have any queries?

